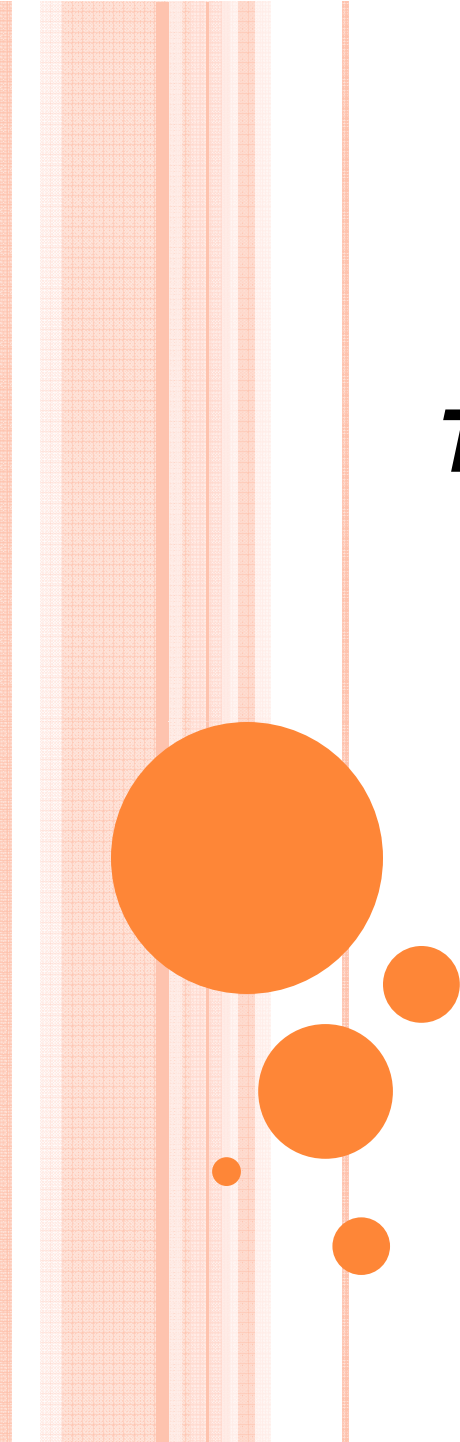
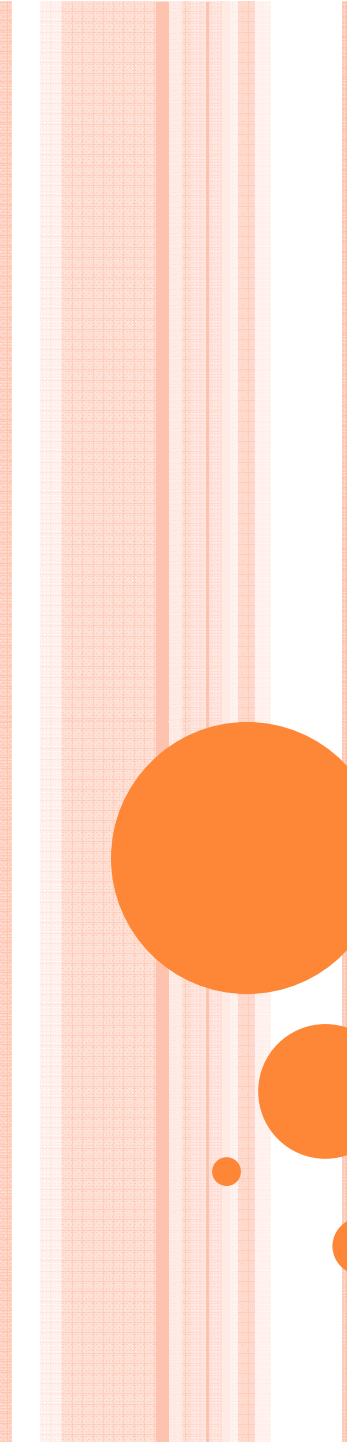
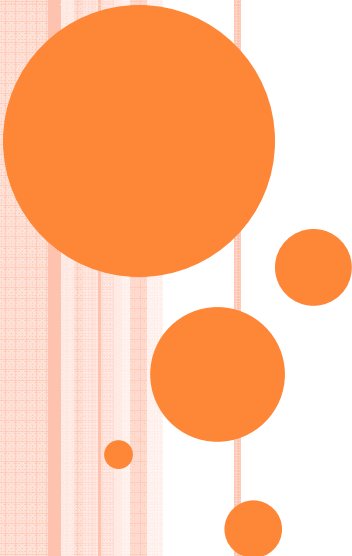




Transducer & Positioner Basics



I/P TRANSDUCERS



TRANSDUCER

○ Definition

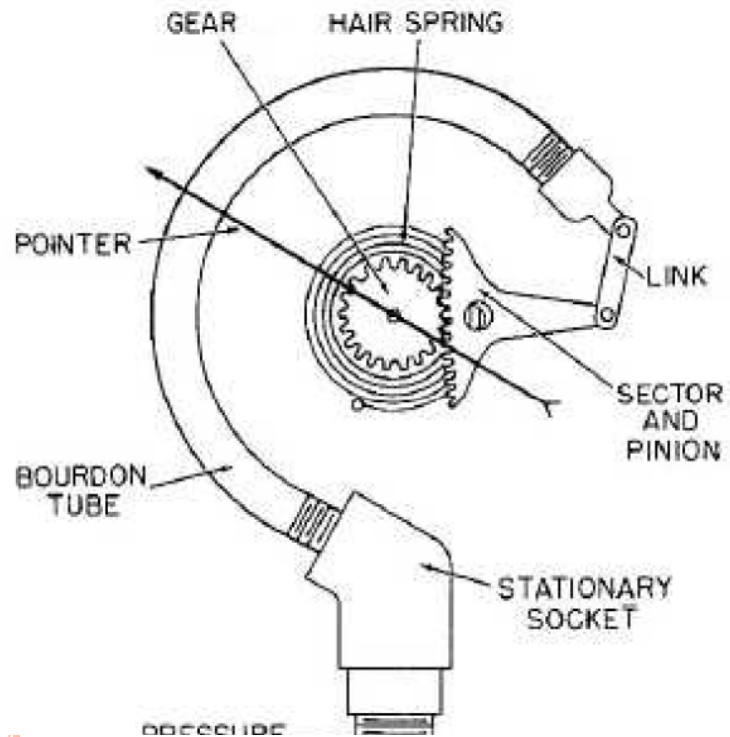
- A device that converts variations in a physical quantity (such as pressure or temperature) into a measurable signal, preferably an electrical signal

○ Classification

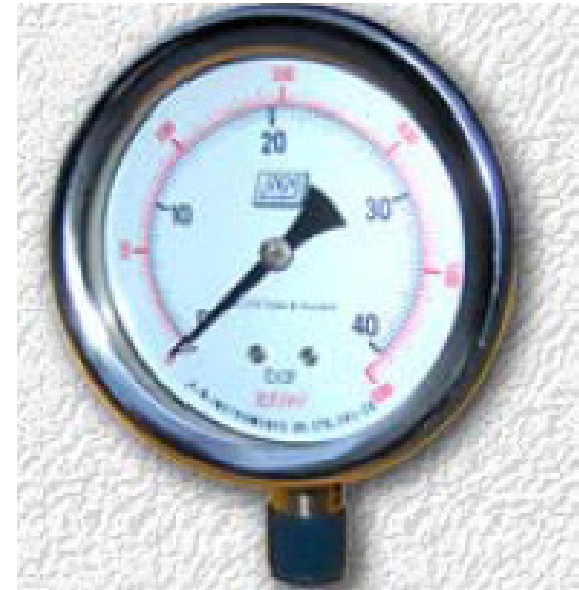
- Mechanical
- Electrical/ Electro Mechanical
- Pneumatic
- Electro Pneumatic



MECHANICAL

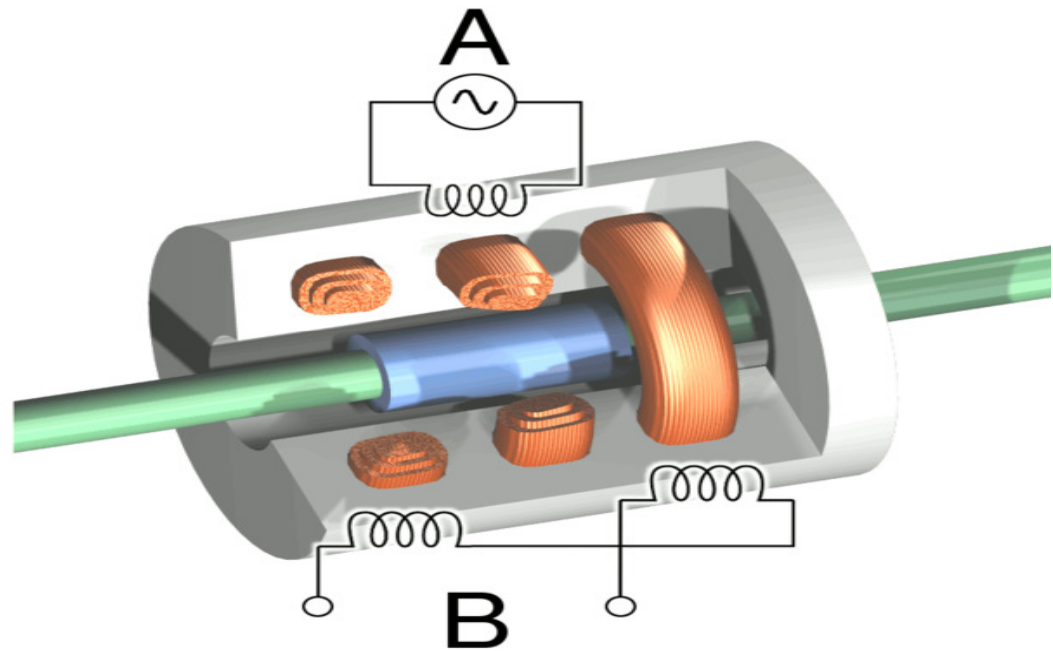
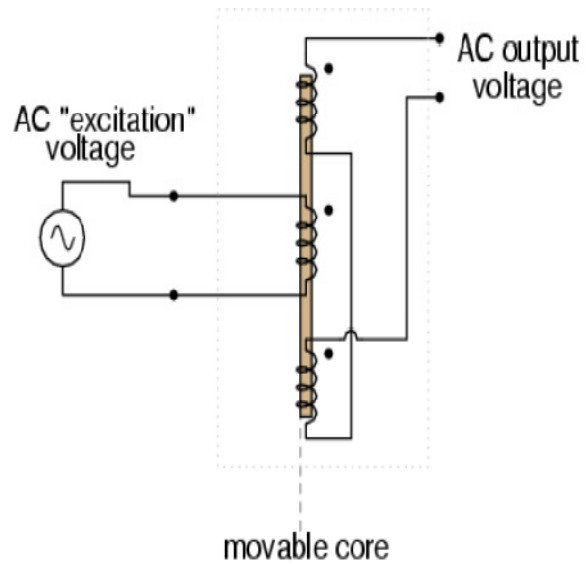


Bourdon Tube



ELECTRO-MECHANICAL

*The Linear Variable Differential Transformer
(LVDT)*



PNEUMATIC

- **Pneumatic Sensors**
- **Works in conjunction with Electric/Electronic/Mechanical Components**



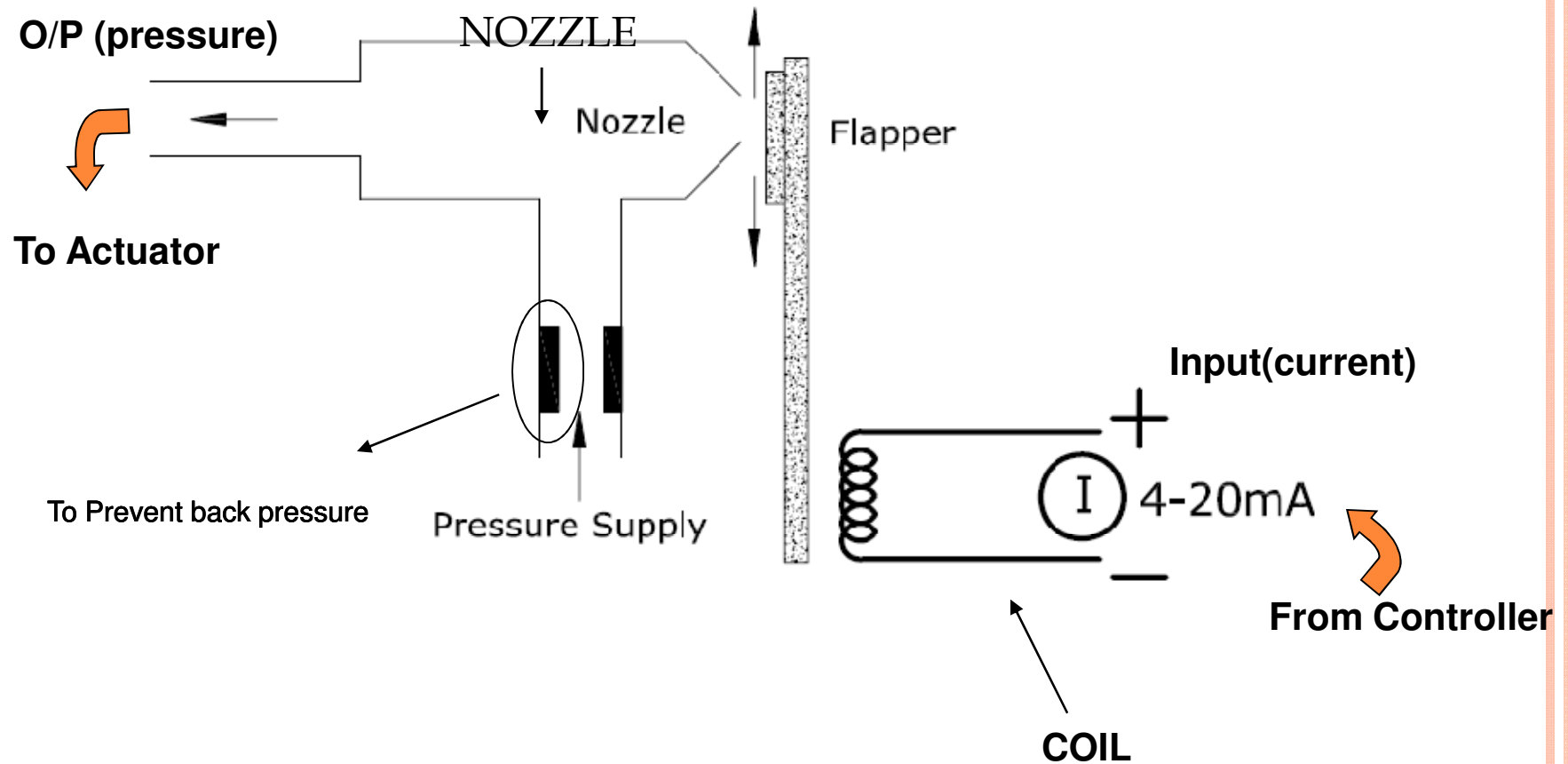
Diaphragm



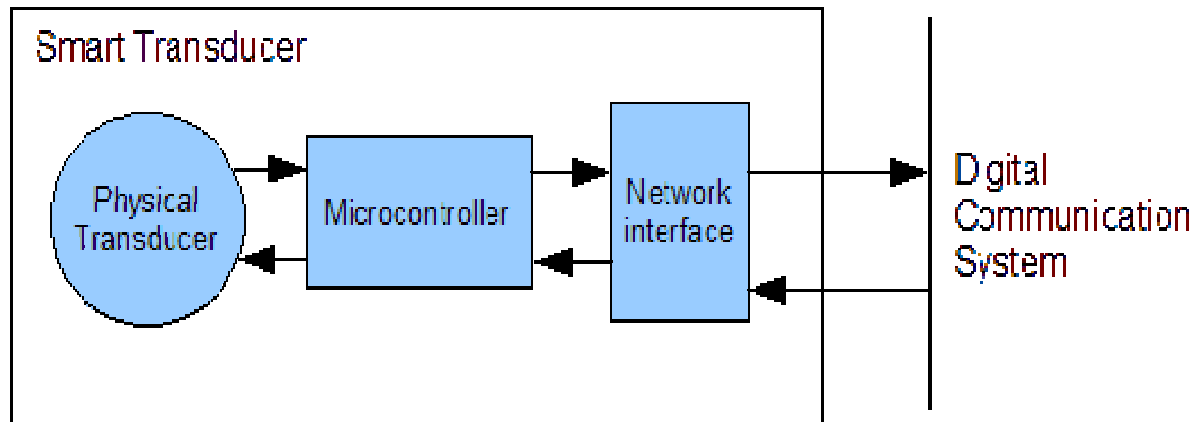
Bellows



ELECTRO - PNEUMATIC (I/P TRANSDUCER)



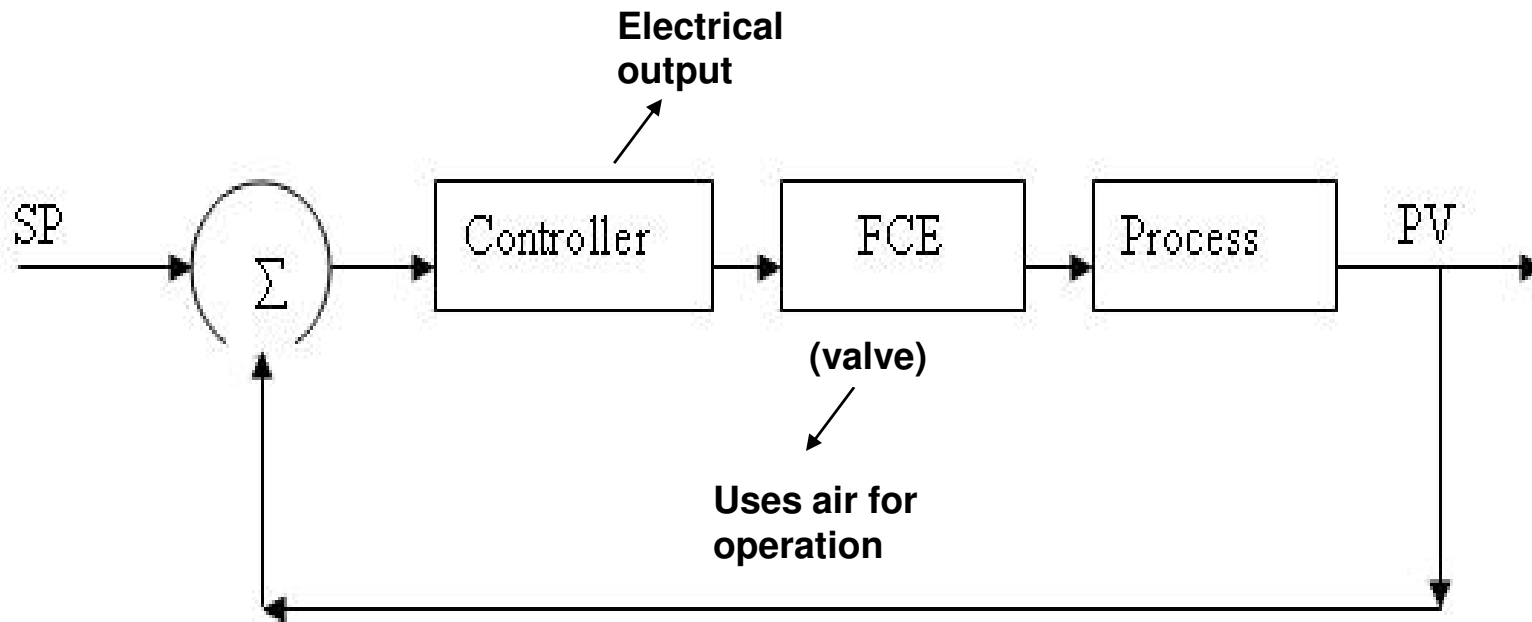
SMART TRANSDUCER



- The integration of an analog or digital sensor, a processing unit, and a communication interface is referred to as smart Transducer.
- Output can be read, calibrated, configured with the use of HART master (handheld communicator) from any junction.

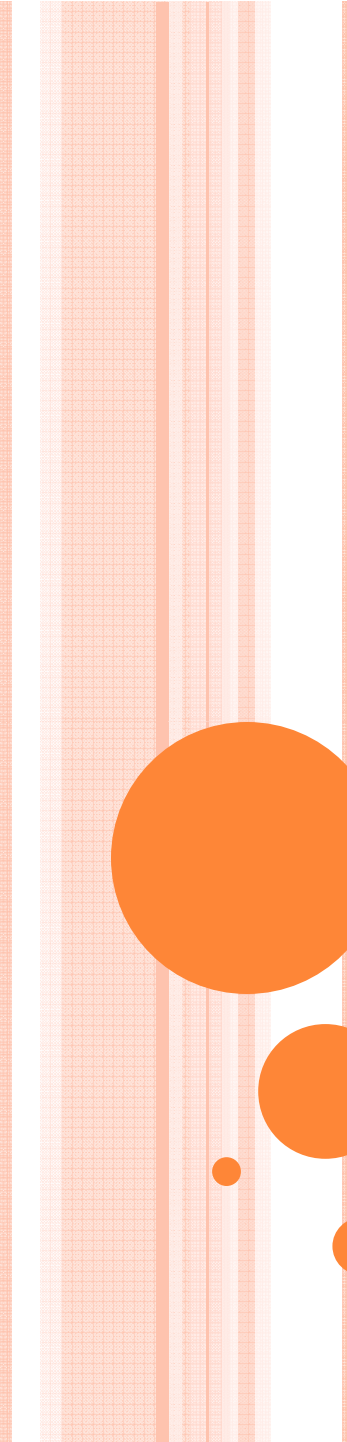


NEED FOR I/P TRANSDUCER IN VALVE OPERATION



Electrical Control Signal needs to be converted into a Pneumatic signal through an I/P Transducer





I/P POSITIONERS

I/P POSITIONERS

○ **Role of Positioner**

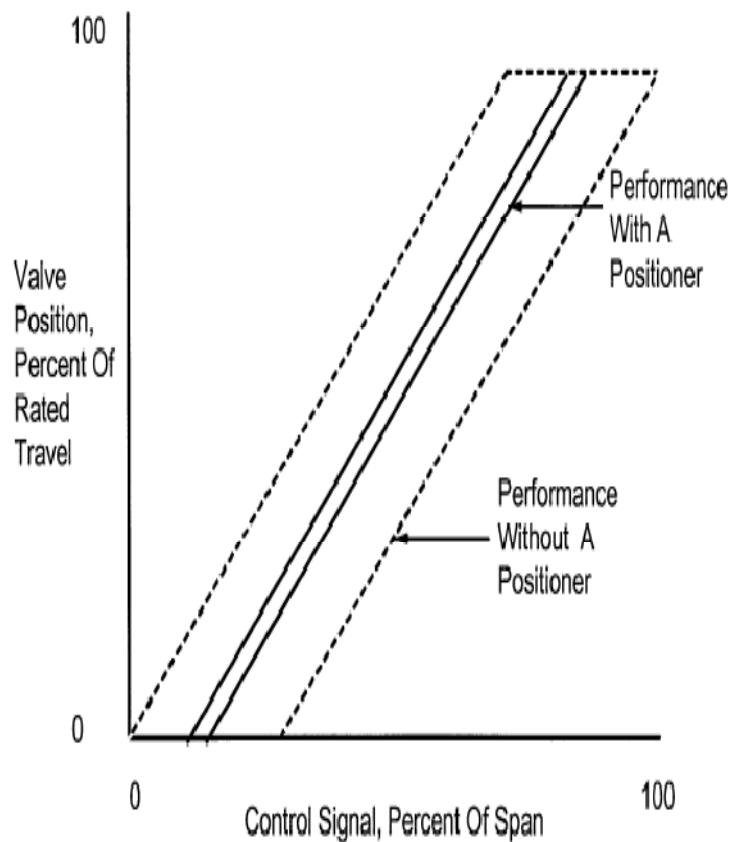
- A device that keeps the valve stem at desired position with greater accuracy and stability
- Provides feedback to the control system
- Reduces dead band and hysteresis in the system

○ **Pneumatic Positioners**

- Used in conventional pneumatic plants.
- Input signal → 3 – 15 psig
- No advanced diagnostics/performance diagnostics features.
- Mainly used in MRO business now.



ROLE OF POSITIONER



- Friction- Leading cause of imprecise valve stem positioning.
- Friction from packing, seals and bearings can cause the valve stem or shaft to stick in a fixed position even though the actuator loading pressure is changing.



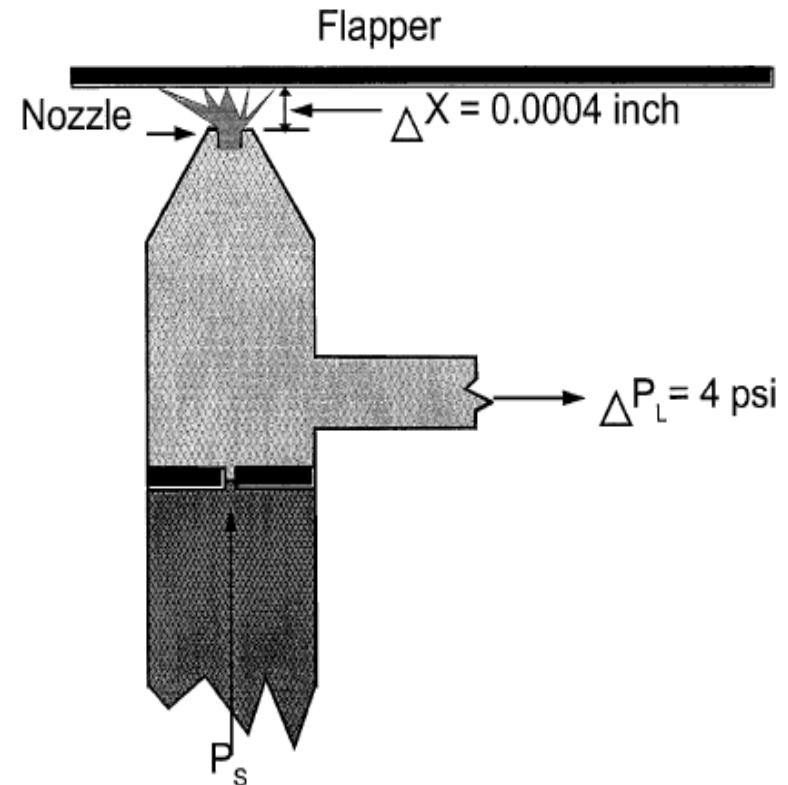
FLAPPER NOZZLE

Advantages

- High gain

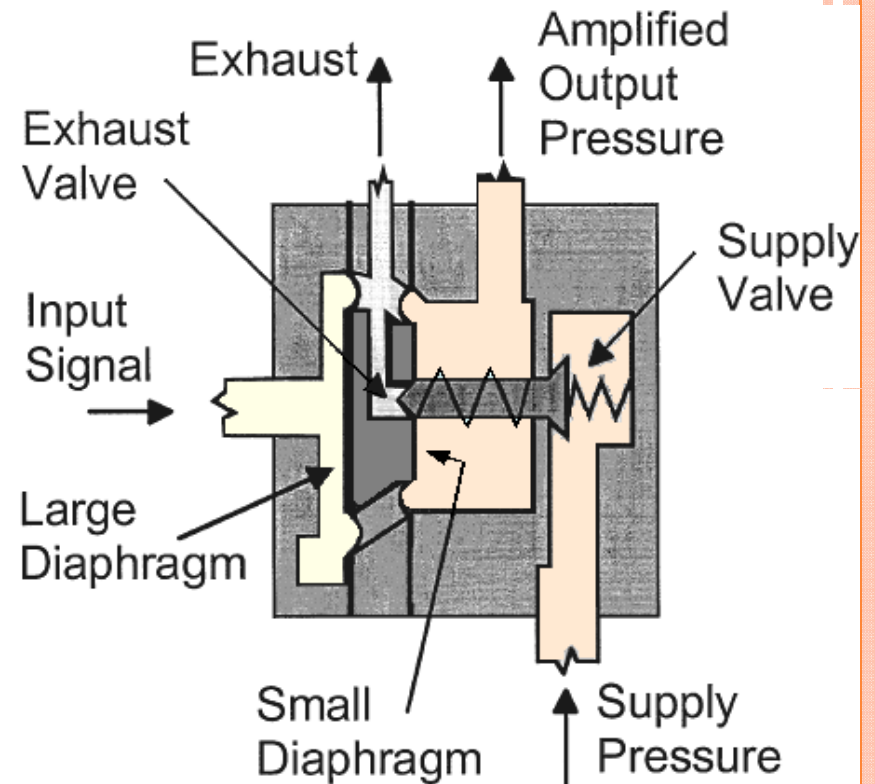
Operation

- Nozzle has an internal fixed restriction and a variable restriction at the end.
- Flapper moves closer then back pressure generated increases and vice versa.



RELAY OPERATION

- Pressure amplification.
- Based on input signal, supply valve opens.
- Exhaust closes.
- Supply pressure flows to the actuator proportional to the input signal.

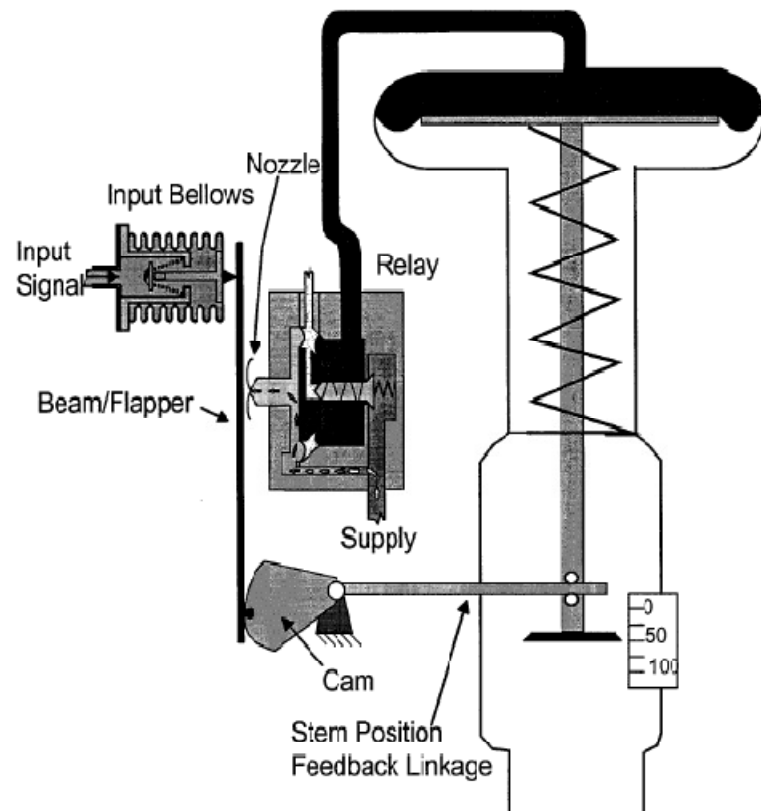


VARIOUS POSITIONERS

- For Sliding Stem Valves
 - Single Acting Positioners
 - 3582, 3660, 3582i, 3661, 3611J
 - Double Acting Positioners
 - 3611JP
- For Rotary Valves
 - Single Acting Positioners
 - 3610J, 3620J
 - Double Acting Positioners
 - 3610JP, 3620JP



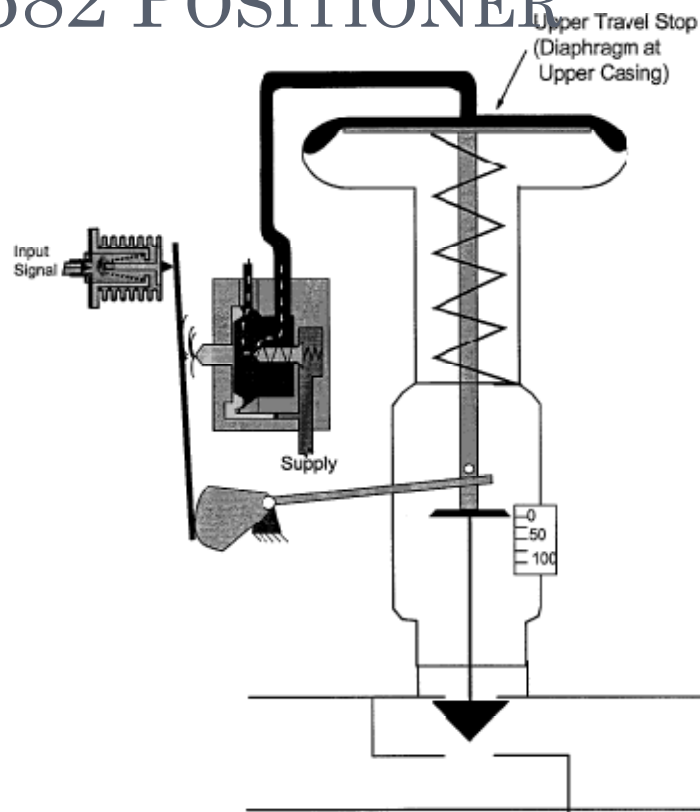
3582 POSITIONER



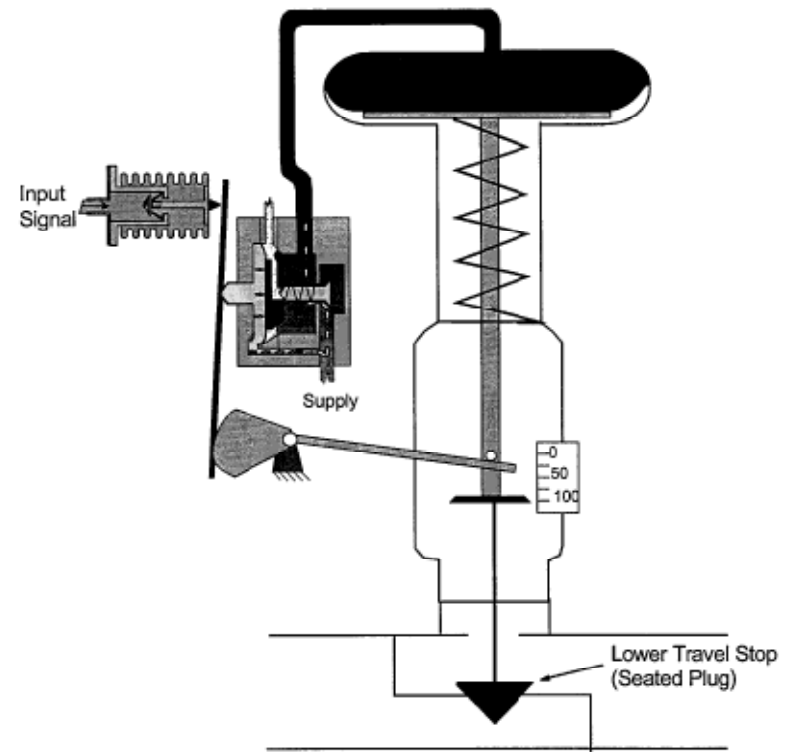
- Includes Flapper-Nozzle assembly
- Relay used for pressure amplification
- Beam is balanced between the Input signal and the travel feedback linkage
- Direct/Reverse acting



3582 POSITIONER



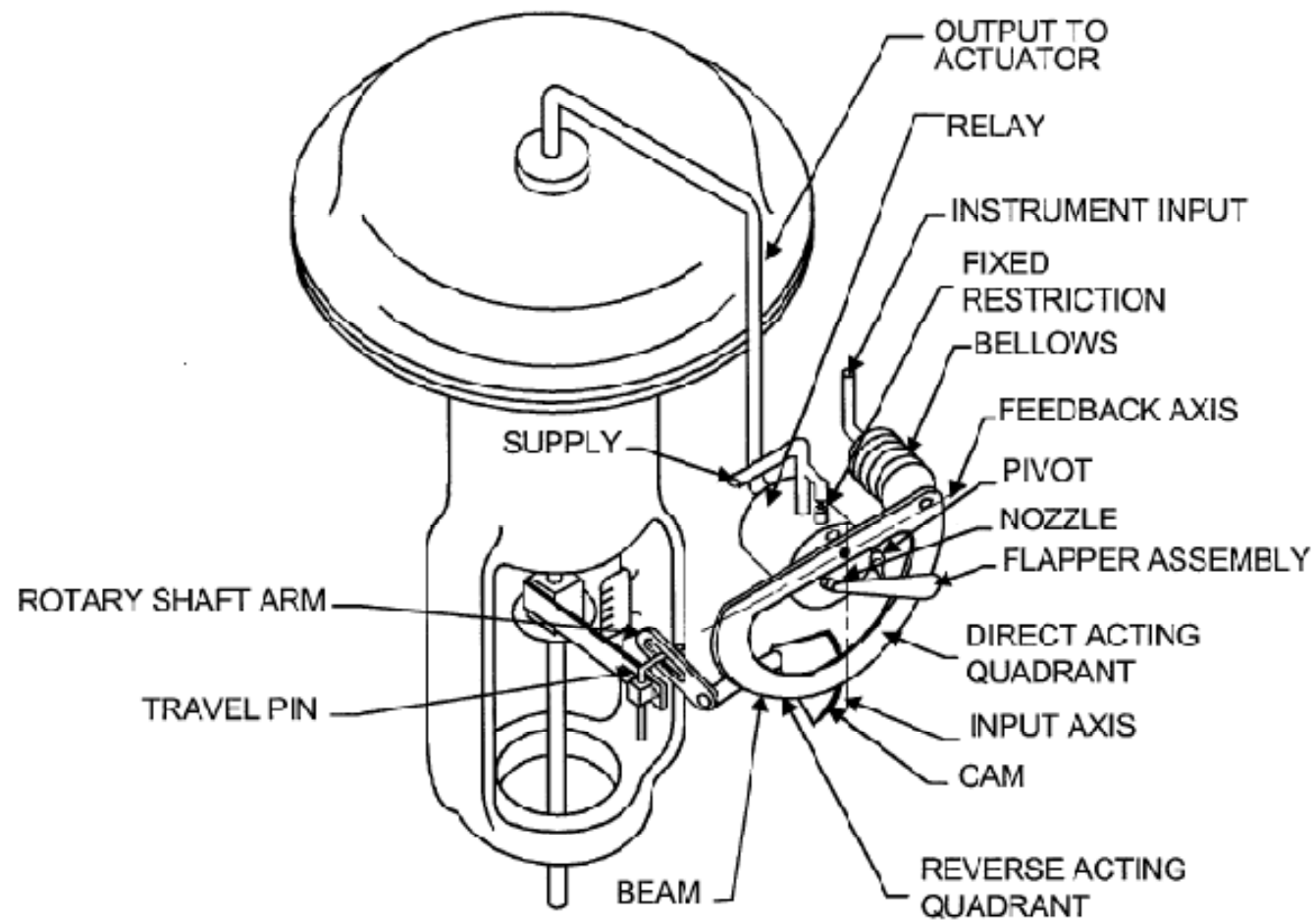
Saturation at low end of scale



Saturation at high end of scale



3582 POSITIONER

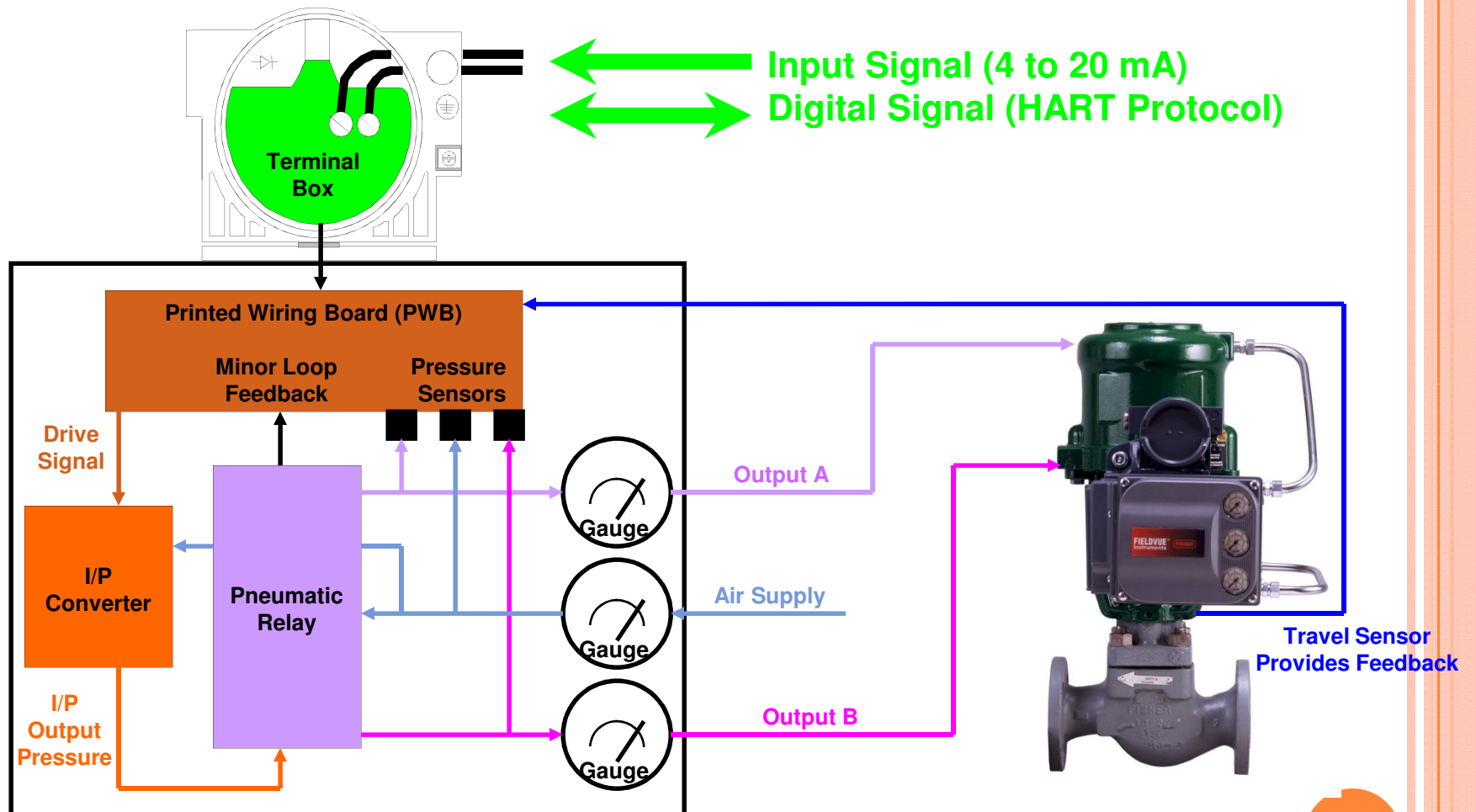


DIGITAL VALVE CONTROLLERS

- High performance I/P positioners
- Features:
 - Stem positioning accuracy
 - Speed and stability
 - Common configuration interface
- Extended features such as
 - Configurable operating parameters
 - Configurable alerts
 - Diagnostics

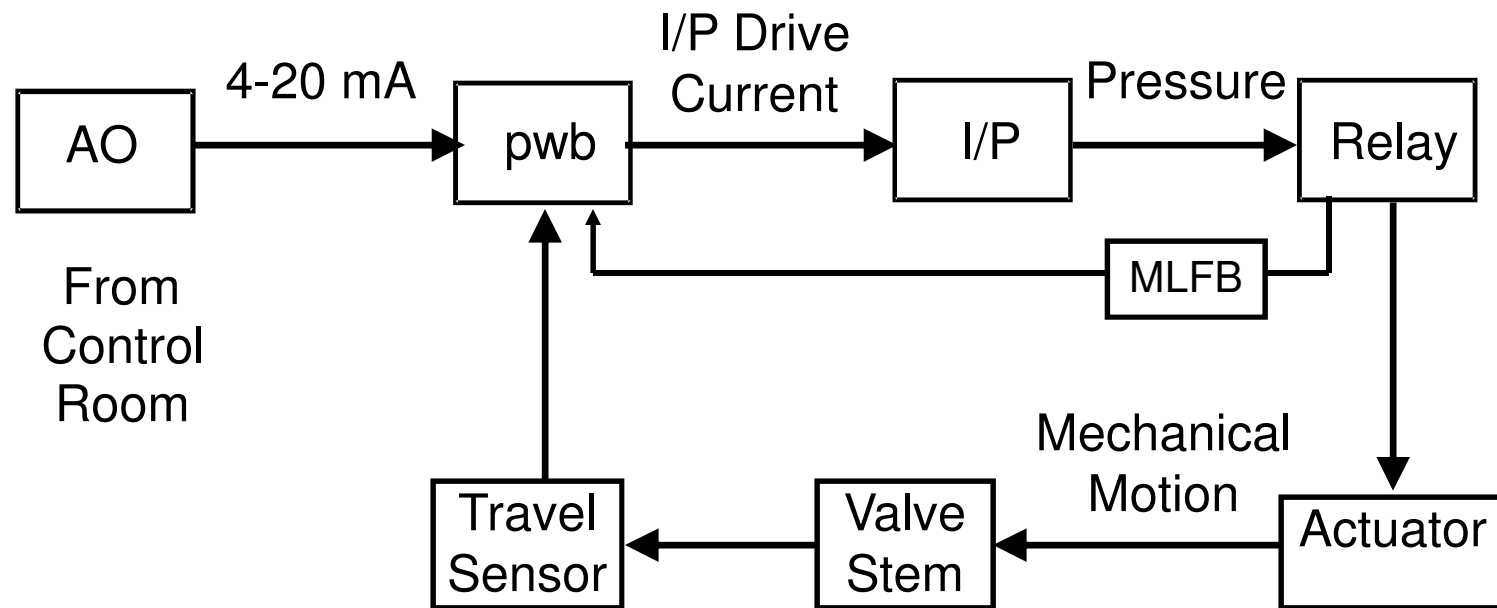


DVC6000 Basic Operation



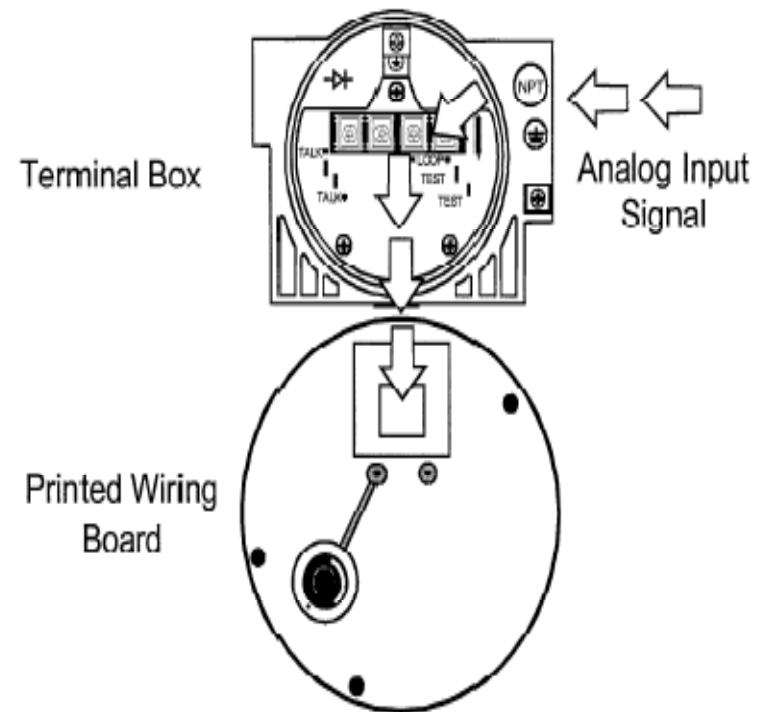
End Build

SYSTEM BLOCK DIAGRAM



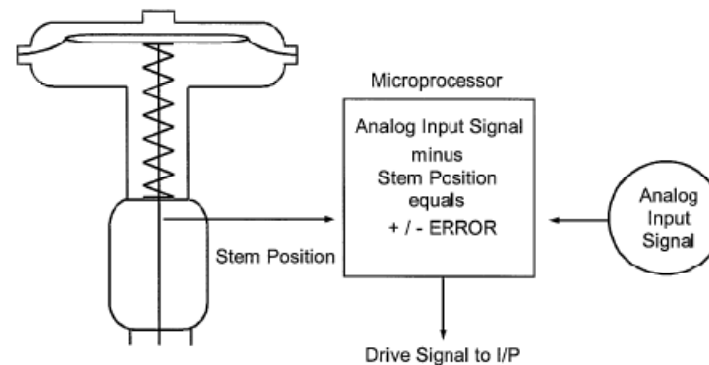
THEORY OF DVC OPERATION

- Analog Signal Path
 - The analog signal, typically 4-20 mA is routed to terminal box through single pair of twisted wires to printed wiring board where the microprocessor reads the signal



THEORY OF DVC OPERATION

- Microprocessor Unit and drive signal
 - Microprocessor compares analog input signal from the loop controller with the stem position signal
 - Stem position signal is generated by potentiometer present within DVC which is connected to valve shaft with feedback linkage



THEORY OF DVC OPERATION

- Drive signal/ I/P converter
 - The error signal generated due to the difference between the input signal and the current position of the stem increases or decreases
 - The change alters the magnetic attraction between the core and the armature of I/P Converter
 - The I/P flapper attached to the armature changes its position causing backpressure alteration

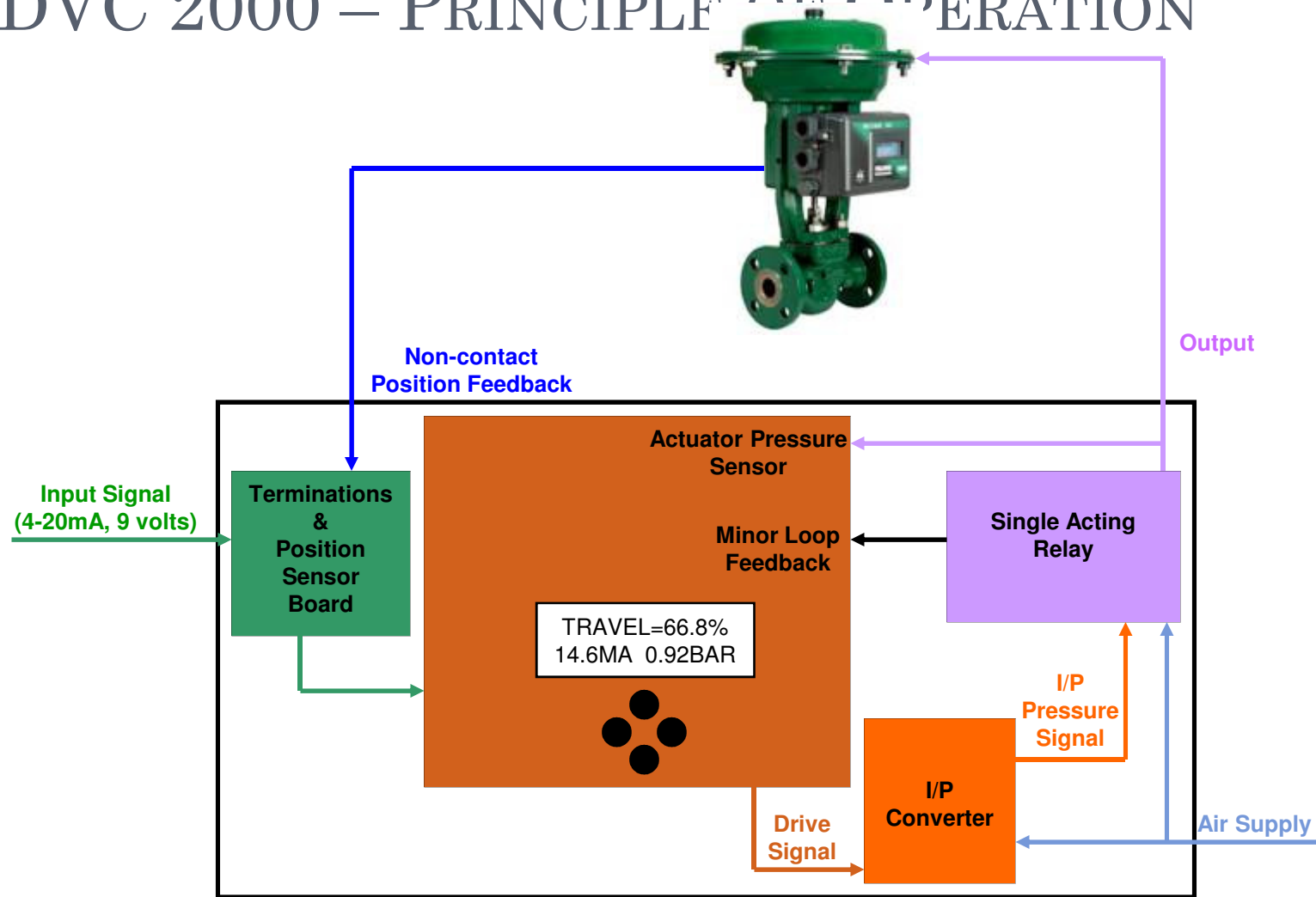


DVC 2000

- Small
- Intrinsically Safe
- Single acting
- Linkage-less, non-contact, position feedback
- Local User Interface (LCD and pushbuttons)
- Multi-lingual (English, French, German, Italian, Spanish, Chinese, Japanese)
- Autotuner (tuning based on actuator size)
- Diagnostic tiers (AC, HC, AD, PD)
- Communication tiers (none, HART, Fieldbus-future)
- NAMUR mounting
- High performance
- Complimentary to the DVC6000 Series
- Limit Switches (2)
- 4-20mA Position Transmitter



DVC 2000 – PRINCIPLE OF OPERATION



DVC FEATURES

- Networking Features
 - HART-protocol communicating .
 - Multiplexing-valve link, DCS.
 - Remote
- Valve link/AMS Diagnostics
 - Valve signature test
 - Dynamic error band test
 - Drive signal test
 - Output signal test
 - Step response test



DVC FEATURES

Network Configuration

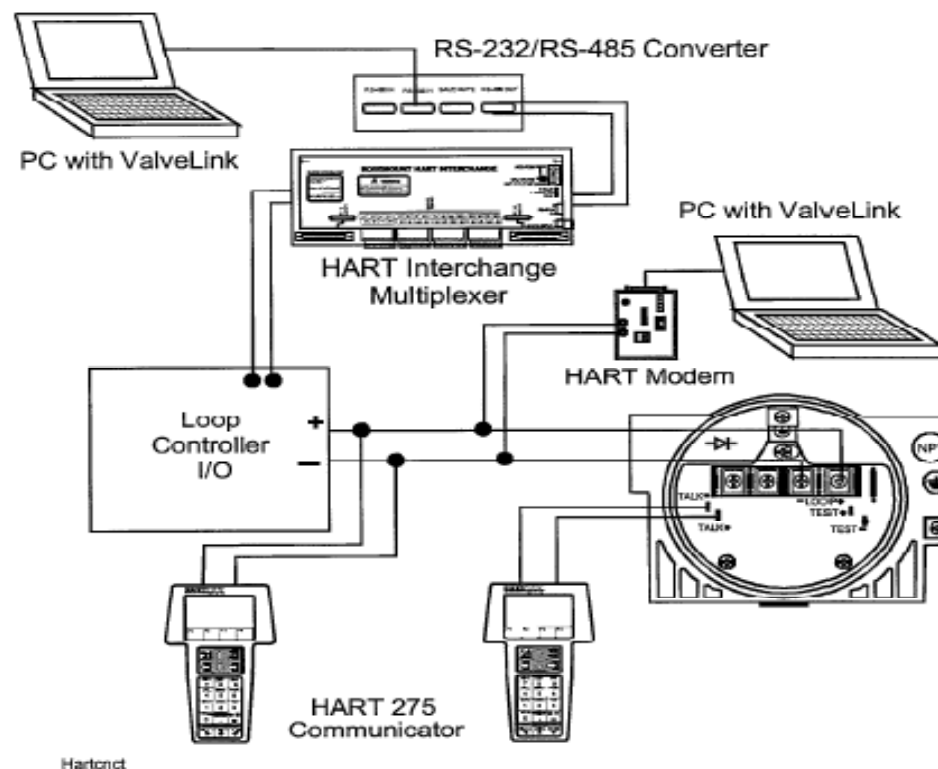


Figure 16-17. Examples Of FIELDVUE Instrument Communication Strategies

EDUCATIONAL SERVICES

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FISHER-ROSEMOUNT

Managing The Process Better.

Thanks

